CHAPTER 3

Pinniped Evolution and Systematics

3.1 INTRODUCTION

Pinnipeds are aquatic members of the mammalian order Carnivora and comprise three monophyletic families: the Otariidae (eared seals or the fur seals and sea lions); the Odobenidae (walruses); and the Phocidae (true or earless seals). Pinnipeds make up slightly more than one-fourth (28%) of the diversity of marine mammals. Thirty-three living species of pinnipeds are distributed throughout the world; 18 phocids, 14 otariids, and the walrus. Roughly, 60% of the combined global population numbers of pinnipeds are phocids; the remaining 40% are otariids and odobenids (Berta and Churchill, 2012). The fossil record indicates that extant pinnipeds represent only a small fraction of what was once a much more diverse group. For example, only a single species of walrus exists today, whereas no less than 14 genera and 15 species existed in the past (Boessenecker and Churchill, 2013). The earliest well-documented record of pinnipeds is from the late Oligocene (27–25 Ma; Figure 3.1), although a slightly earlier record (29 Ma) is suggested, though less well substantiated (Koretsky and Sanders, 2002).

New discoveries of fossil pinnipeds, together with comparative studies of living taxa, have enabled a relatively complete understanding of pinniped origin, diversification, and morphology, though work, of course, remains. The current understanding of these topics is explored in this chapter. Characters defining major groups of pinnipeds are also listed for reference. Controversies regarding the relationship of pinnipeds to other carnivores, relationships among pinnipeds, and the alliance of an extinct pinniped group, desmatophocids, are also considered.

3.2 ORIGIN AND EVOLUTION

3.2.1 Pinnipeds Defined

The name pinniped comes from the Latin *pinna* and *pedis* meaning "feather-footed," referring to the paddle-like fore- and hindlimbs of seals, sea lions, and walruses. Pinnipeds have a basic dichotomy in their lives, in that they spend considerable amounts of time in the water, but also come onto land or ice routinely. In this way, they differ from cetaceans and sirenians, which have become entirely aquatic. In addition to blubber, some pinnipeds also have a thick covering of fur.

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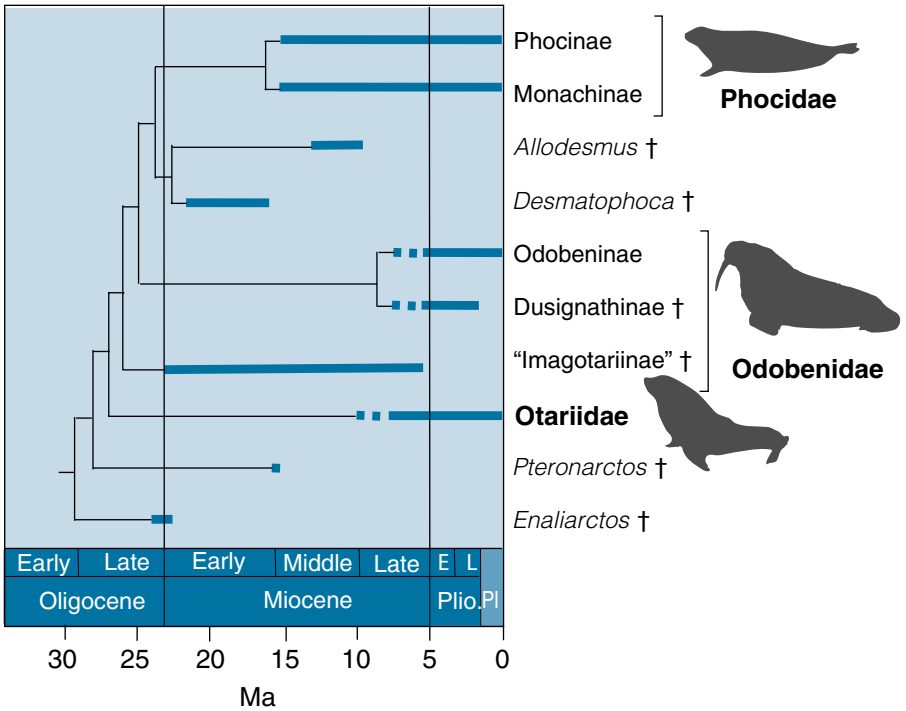

FIGURE 3.1
Chronological ranges of extinct and living pinnipeds. Ma = million years ago.

In seeking the origin of pinnipeds we must first define them. Is the group monophyletic or not? Although this question has been the subject of considerable controversy during the last century (e.g., Uhen, 2007), the majority of scientists today agree that the Pinnipedia represent a natural, monophyletic group. Pinnipeds are diagnosed by a suite of derived morphological characters (for a complete list, see Wyss, 1987, 1988; Berta and Wyss, 1994). All pinnipeds, including both fossil and recent taxa, possess these characters, which are described later, although some of these characters have been modified or lost secondarily in later diverging taxa.

Some of the well-known synapomorphies possessed by pinnipeds (enumerated in Figure 3.2 and illustrated in Figures 3.3–3.5) are defined as follows:

1. *Large infraorbital foramen.* The infraorbital foramen, as the name indicates, is located below the eye socket (or orbit). This opening allows for the passage of blood vessels and nerves that serve the eye and surrounding tissues. It is large in pinnipeds in contrast to its small size in most terrestrial carnivores.
2. *Maxilla makes a significant contribution to the orbital wall.* Pinnipeds display a unique condition among carnivores in which the maxilla (upper jaw) forms part of the lateral and anterior walls of the orbit of the eye. In terrestrial carnivores, the maxilla is usually limited in its posterior extent by contact with several of the facial bones (jugal, palatine, or lacrimal).

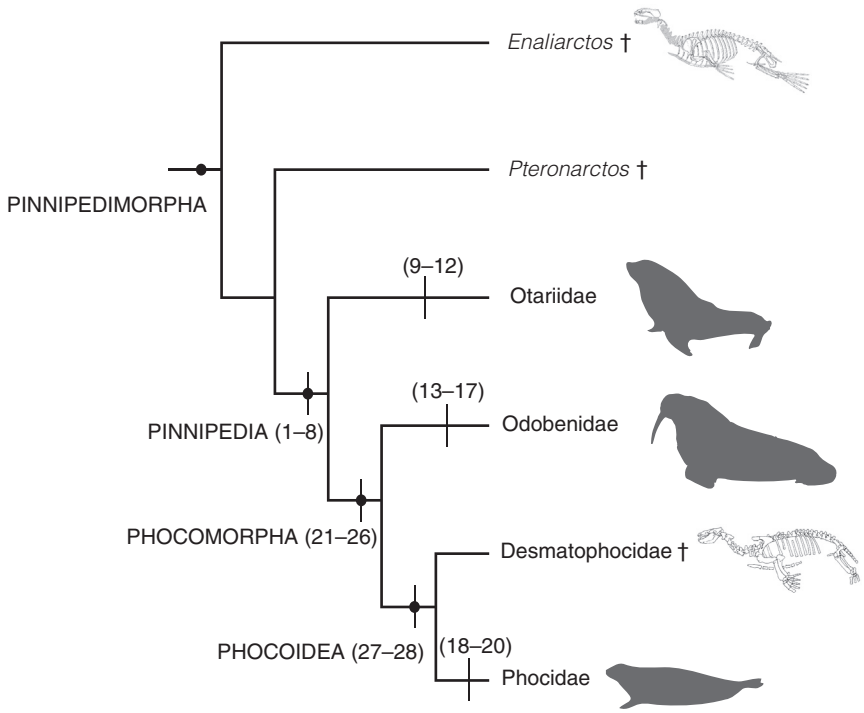

FIGURE 3.2
A cladogram depicting the relationships of the major clades of pinnipeds. Numbers at nodes refer to synapomorphies listed in the text and † = extinct taxa; see also Figures 3.3–3.5. For more detailed cladograms of individual families, see Figures 3.14, 3.19, and 3.23. Modified from Wyss (1988), Berta and Wyss (1994), Deméré (1994b).

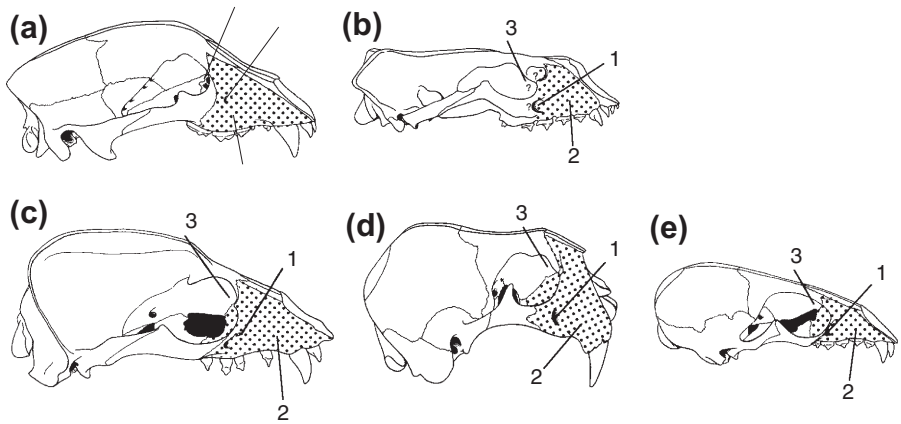

FIGURE 3.3
Lateral views of the skulls of representative pinnipeds and a generalized terrestrial arctoid. (a) bear, *Ursus americanus*; (b) fossil pinnipedimorph, *Enaliarctos mealsi*, (c) otariid, *Zalophus californianus*; (d) walrus, *Odobenus rosmarus*; (e) phocid, *Neomonachus schauinslandi*, illustrating pinniped synapomorphies. Character numbers (see text for further description): 1 = large infraorbital foramen; 2 = maxilla (stippled) makes a significant contribution to the orbital wall; 3 = lacrimal absent or fused and does not make contact with the jugal. From Berta and Wyss (1994).

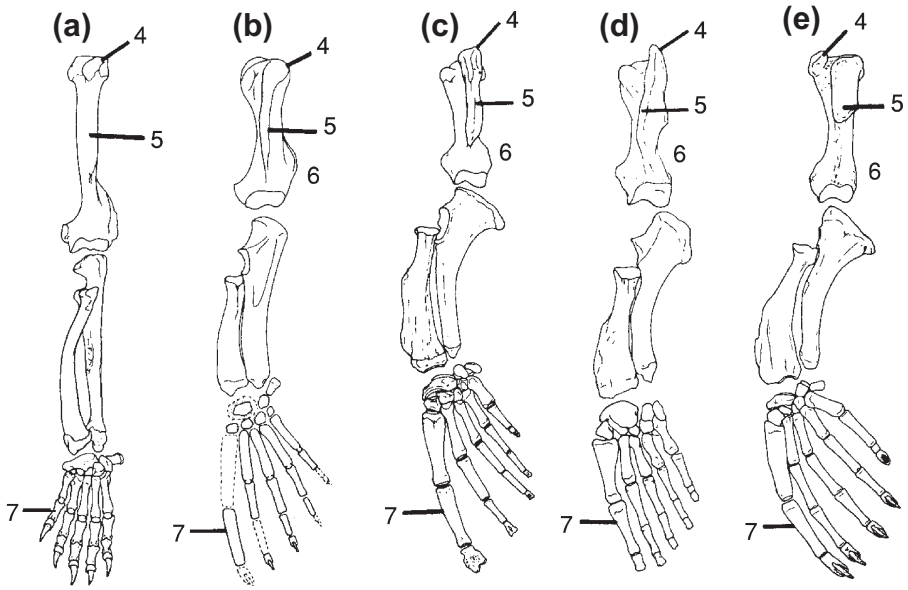

FIGURE 3.4
Left forelimbs of (a) a generalized terrestrial arctoid and (b–e) representative pinnipeds in dorsal view, illustrating pinniped synapomorphies. Labels as in Figure 3.3 plus character numbers (see text for further description): 4 = greater and lesser humeral tubercles enlarged; 5 = deltopectoral crest of humerus strongly developed; 6 = short, robust humerus; 7 = digit I on manus emphasized. From Berta and Wyss (1994).

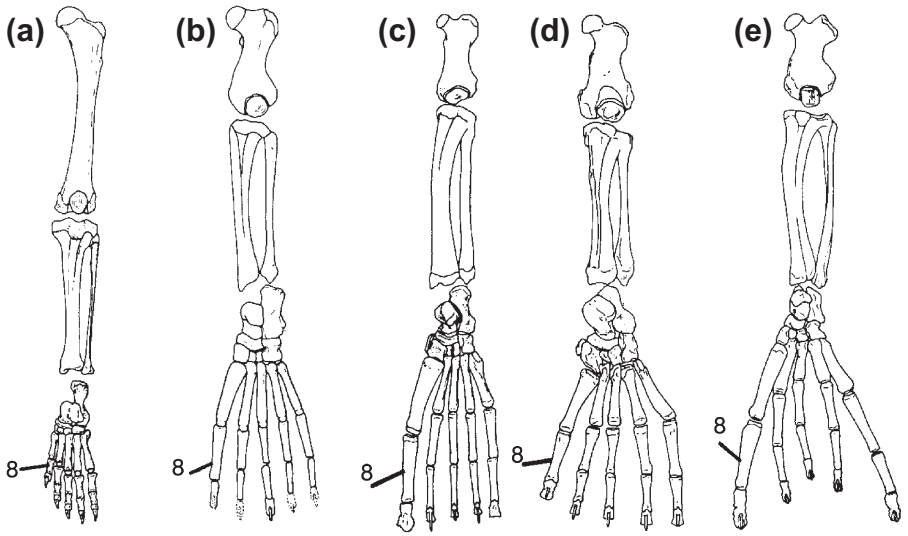

FIGURE 3.5
Left hindlimbs of (a) a generalized terrestrial arctoid and (b–e) representative pinnipeds in dorsal view illustrating pinniped synapomorphies. Labels as in Figure 3.3. Character number (see text for further description): 8 = digits I and V on the foot are emphasized. From Berta and Wyss (1994).

3. *Lacrimal absent or fusing early in ontogeny and does not contact the jugal.* Associated with the pinniped configuration of the maxilla (character 2), there is a great reduction or absence of one of the facial bones, the lacrimal. Terrestrial carnivores have a lacrimal bone that contacts the jugal or is separated from it by a thin sliver of the maxilla.
4. *Greater and lesser humeral tubercles enlarged.* Pinnipeds are distinguished from terrestrial carnivores by having strongly developed tubercles (rounded prominences) on the proximal end of the humerus (upper arm bone).
5. *Deltopectoral crest of humerus strongly developed.* The crest on the shaft of the humerus for insertion of the deltopectoral muscles in pinnipeds is strongly developed in contrast to the weak development observed in terrestrial carnivores.
6. *Short and robust humerus.* The short and robust humerus of pinnipeds is in contrast to the long, slender humerus of terrestrial carnivores.
7. *Digit I on the hand emphasized.* In the hand of pinnipeds the first digit (thumb equivalent) is elongated, whereas in other carnivores the central digits are the most strongly developed.
8. *Digits I and V on the foot emphasized.* Pinnipeds have elongated side toes (digits I and V, equivalent to the big toe and little toe) of the foot, whereas in other carnivores the central digits are the most strongly developed.

3.2.2 Pinniped Affinities

Since the name Pinnipedia was first proposed by Illiger in 1811, there has been debate on the relationships of pinnipeds to one another and to other mammals (reviewed in Uhen, 2007). Two hypotheses have been proposed. A monophyletic hypothesis proposes that the three pinniped families share a single common evolutionary origin (Figure 3.6(a)). A **diphyletic** view (also referred to as pinniped diphyly; Figure 3.6(b)) calls for the origin of pinnipeds from two carnivore lineages, the alliance of odobenids and otariids being somewhere near ursids (bears) and a separate origin for phocids from the mustelids (weasels, skunks, otters, and their kin).

Traditionally, morphological and paleontological evidence supported pinniped diphyly (e.g., Tedford, 1976; Repenning et al. 1979; Muizon, 1982). Nevertheless, on the basis of his reevaluation of the morphological evidence, Wyss (1987) argued in favor of a return to the single origin interpretation. This hypothesis of pinniped monophyly has received considerable support from both morphological (e.g., Wyss and Flynn, 1993; Berta and Wyss, 1994) and molecular studies (e.g., Higdon et al. 2007; Fulton and Strobeck, 2010; Nyakatura and Bininda-Emonds, 2012).

All recent workers agree that the closest relatives of pinnipeds are arctoid carnivores, which include procyonids (raccoons and their allies), mustelids, and ursids, although the specific arctoid group that forms the closest alliance with pinnipeds is still disputed. There is evidence to support a mustelid (Bininda-Emonds and Russell, 1996; Flynn and Nedbal, 1998; Bininda-Emonds et al. 1999; Fulton and Strobeck, 2006; Nyakatura and Bininda-Emonds, 2012), an ursid (Wyss and Flynn, 1993; Berta and Wyss, 1994), or an ursid–mustelid (Davis et al. 2004) ancestry.

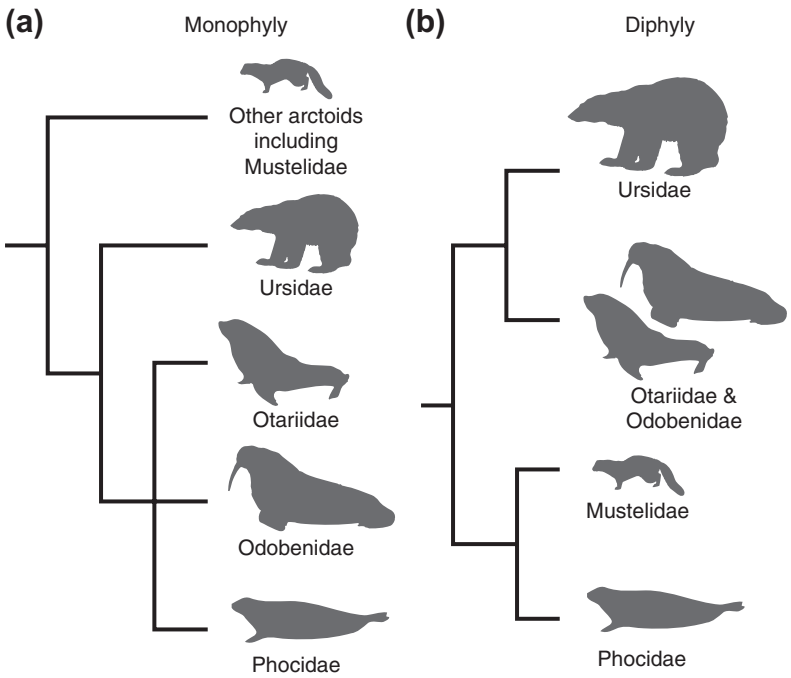

FIGURE 3.6
Alternative hypotheses for relationships among pinnipeds (a) monophyly with ursids as the closest pinniped relatives, and (b) diphyly in which phocids and mustelids are united as sister taxa, as are otariids, odobenids, and ursids.

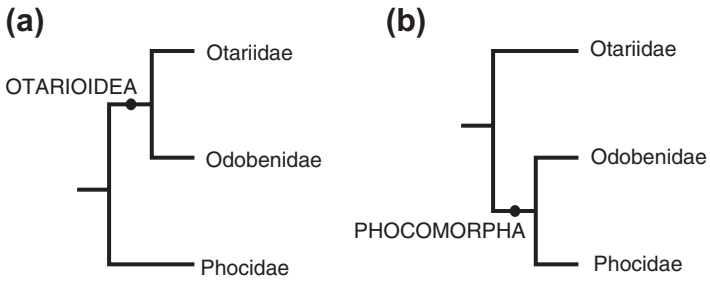

FIGURE 3.7
Alternative hypotheses for the phylogenetic position of the walrus: (a) "Otarioidea" clade or (b) Phocomorpha clade.

Although both morphological and molecular data support pinniped monophyly, disagreement still occurs regarding the relationships among pinnipeds. Most of the controversy lies in the debate as to whether the walrus is most closely related to phocids or to otariids. Some morphologic evidence from extant pinnipeds unites the walrus and the phocids as sister groups (Figure 3.7(a); Wyss, 1987; Wyss and Flynn, 1993; Berta and Wyss, 1994), which is discussed later in this chapter. An alternative view based on strong and consistent support from

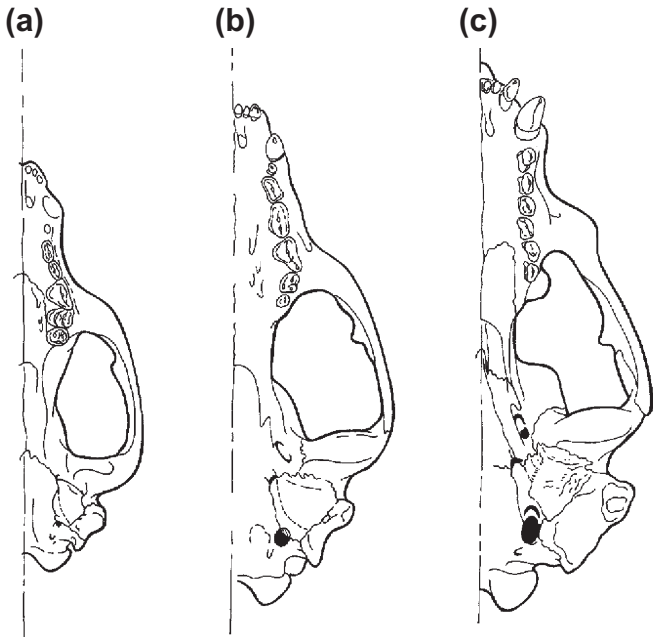

FIGURE 3.8
Skulls and dentitions of representative pinnipeds and a generalized terrestrial arctoid in ventral view. (a) Stem bear, *Pachcynodon* (Oligocene, France); (b) stem pinnipedimorph, *Enaliarctos mealsi* (early Miocene); and (c) crown otariid, *Arctocephalus* (Holocene, South Atlantic). From Tedford (1976).

molecular data (e.g., Davis et al. 2004; Arnason et al. 2006; Higdon et al. 2007), with some support from combined analyses (e.g., Flynn and Nedbal, 1998), supports an alliance between the walrus and the otariids (Figure 3.7(b)).

3.2.3 Stem Pinnipeds

An understanding of the evolution of stem pinnipeds necessitates knowledge of certain fossil taxa. The earliest diverging lineage of pinnipeds were members of the Pinnipedimorpha clade, which appear to have originated in the eastern North Pacific (Oregon) during the late Oligocene (27–25 Ma; see Figure 3.1). The earliest well-represented pinnipedimorph, *Enaliarctos*, is known by five species (Mitchell and Tedford, 1973; Berta, 1991). The ancestral pinnipedimorph dentition, exemplified by *Enaliarctos barnesi* and *Enaliarctos mealsi*, is **heterodont**, with large blade-like cusps on the upper cheek teeth that were well adapted for shearing (Figure 3.8). These dental features together with those from the skull (when compared with terrestrial carnivores) indicate closest similarities in terms of derived characters with archaic bears (amphicynodonts; see Figure 3.8). The recent discovery of a semiaquatic carnivore, *Pujila darwini*, from 20 to 24 Ma rocks in the Arctic, has been suggested to be either an early pinnipedimorph or an early arctoid, likely a pinniped relative. (Rybczynski et al. 2009; Figure 3.9).

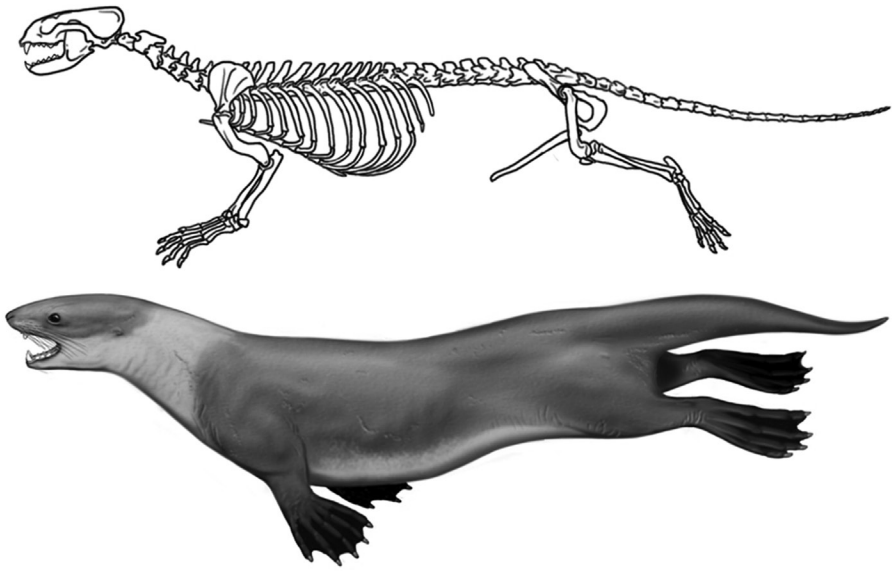

FIGURE 3.9
Skeletal and life restoration of *Pujila darwini*. Illustrated by Carl Buell.

Other species of the genus *Enaliarctos* show a trend toward a decreasing shearing function of the cheek teeth (e.g., reduction in the number and size of cusps). These dental trends herald the development of simple peg-like, or **homodont**, cheek teeth, which are characteristic of most living pinnipeds (Berta, 1991). The latest record of *Enaliarctos* is from rocks of 25 to 18 Ma in age, located along the Oregon coast. An "enaliarctine" pinniped has also been reported from the western North Pacific (Japan) in rocks that are from the Miocene (17.5–17 Ma; Kohno, 1992), although the specimen needs further study before its taxonomic assignment can be confirmed.

The pinnipedimorph *E. mealsi* is represented by a nearly complete skeleton from central California (Figure 3.10; Berta et al. 1989; Berta and Ray, 1990). The entire animal is estimated to have been 1.4–1.5 m in length and between 73 and 88 kg in weight, roughly the size and weight of a small male harbor seal. *E. mealsi* was capable of considerable lateral and vertical movement of the vertebral column. In addition, both its fore- and hindlimbs were modified as flippers and used in aquatic locomotion. Several features of the hindlimb suggest that *E. mealsi* was also capable of maneuvering well on land and probably spent more time near or on shore than extant pinnipeds (see also Chapter 8). Unlike *Enaliarctos* and the pinnipeds, *Puijila* did not possess flippers and more nearly resembled otters in its possession of a long tail and large, probably webbed feet (see Figure 3.9).

A later diverging lineage of fossil pinnipeds, more closely allied with pinnipeds than with *Enaliarctos*, includes *Pteronarctos* and *Pacificotaria* from the early–middle Miocene (19–15 Ma) of coastal Oregon (Barnes, 1989, 1992; Berta, 1994; see Figure 3.1). A striking osteological feature in all pinnipeds is the geometry

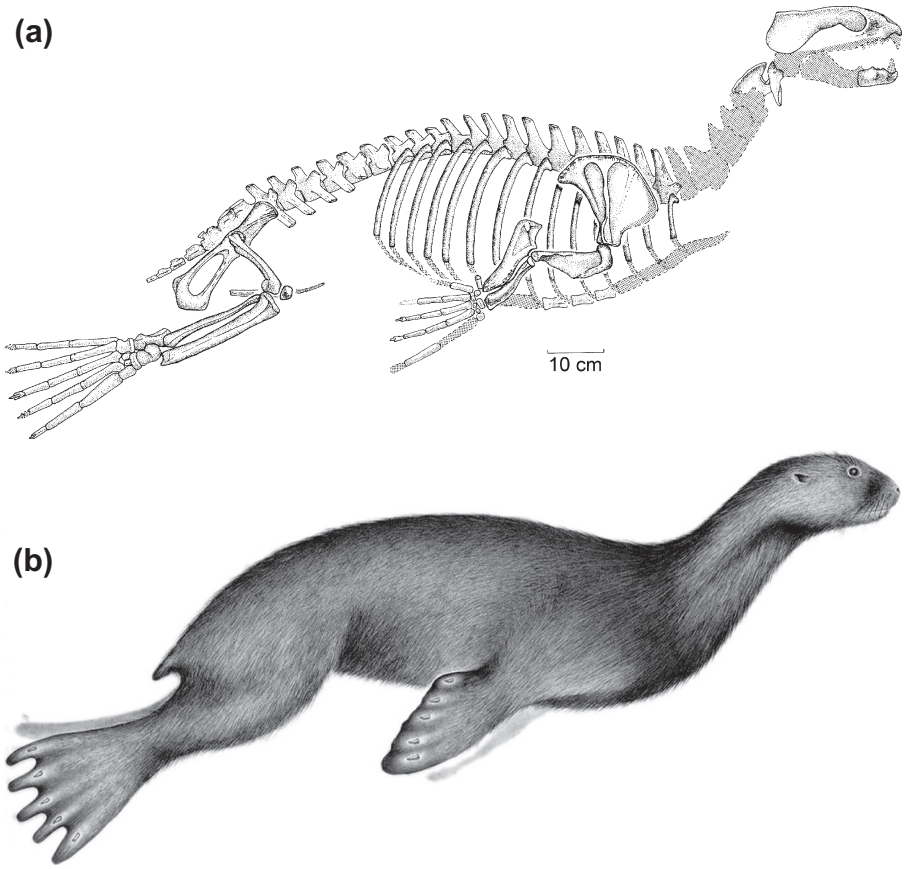

FIGURE 3.10
The pinnipedimorph, *Enaliarctos mealsi* (a) skeletal reconstruction and (b) life restoration. Total estimated length, snout to tail, 1.4–1.5 m. Shaded areas are hypothesized reconstructions of unpreserved bones. From Berta and Ray (1990).

of bones that compose the orbital region (Wyss, 1987). In *Pteronarctos*, the first evidence of the uniquely developed maxilla is seen. In addition, in *Pteronarctos* the lacrimal is greatly reduced or absent, as it is in the pinnipeds. A shallow pit on the palate between the last premolar and the first molar, seen in *Pteronarctos* and pinnipeds, is indicative of a reduced shearing capability of the teeth and marks the beginning of a trend toward homodonty.

3.2.4 Crown Pinnipeds

3.2.4.1 FAMILY OTARIIDAE: SEA LIONS AND FUR SEALS

The otariids are characterized by the presence of external ear flaps, or **pinnae**, and for this reason they are sometimes called eared seals (Figure 3.11). Another characteristic feature of otariids that can be used to distinguish them from phocids is their method of movement on land. Otariids can turn their

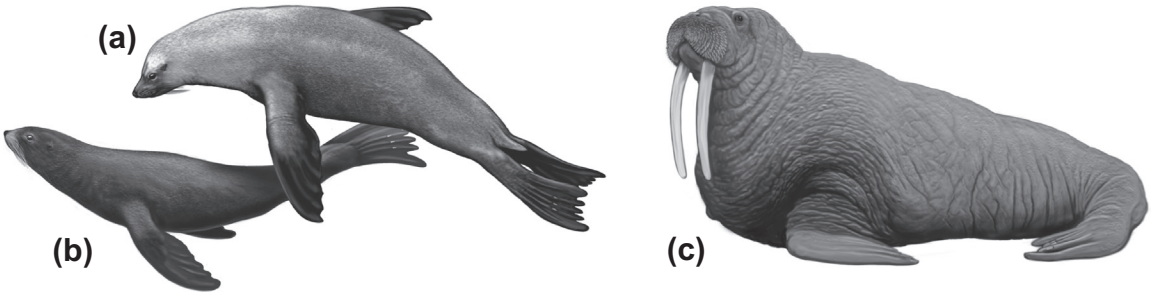

FIGURE 3.11
Representative otariids; (a) California sea lion (*Zalophus californianus*), (b) northern fur seal (*Callorhinus ursinus*), and the odobenid (c) the walrus (*Odobenus rosmarus*). Illustrations by Carl Buell.

hindflippers forward and use them to walk when they are on land (described in more detail in Chapter 8). Otariids are generally smaller than phocids and are shallow divers. Although not exclusive synapomorphies, a few characters used to distinguish otariids relative to other pinniped lineages are the following (Figure 3.12): frontal bones extend anteriorly between the nasal bones; supraorbital processes of the frontal bones are large and shelf-like, especially among adult males; secondary spine subdivides the supraspinous fossa of the scapula; uniformly spaced pelage units; trachea has an anterior bifurcation of the bronchi (Fay, 1981; King, 1983a). Stem otariids include the fossil taxa *Pithanotaria* and *Thalassoleon* spp. Crown otariids (extant taxa and their fossil relatives e.g., *Hydrarctos* and *Proterozetes*) are diagnosed as a monophyletic group by the following dental and osteological characters (Figures 3.2 and 3.12) (Churchill et al. 2014):

9. *Single-rooted P^3 and P^4.* In crown otariids, P^3 and P^4 are single rooted. In stem pinnipeds (*Enaliarctos* and *Pteronarctos*), $P^{3\text{-}4}$ are triple (P^4) or double rooted.
10. *Single-rooted P_2 and P_4.* In crown otariids, P_2 and P_4 are single rooted. In stem pinnipeds (*Enaliarctos* and *Pteronarctos*), P_2 is double rooted. In these stem pinnipeds, P_4 is double rooted. Fusion of tooth roots has occurred multiple times among different pinniped lineages. Fusion of postcanine roots is likely an adaptation to crowd teeth anteriorly in the jaw to increase effective gape (Boessenecker, 2011; Figure 3.13).
11. *Pronator teres insertion positioned on the proximal 40% of the radius.* In crown otariids, the pronator teres process is positioned on the proximal 40% of the radius. In stem otariids (*Pithanotaria*, *Thalassoleon*), the pronator teres process is positioned on the distal 60% of the radius.
12. *Secondary shelf of the sustentaculum of the calcaneum developed as a wide shelf.* In crown otariids, the secondary shelf of the sustentaculum of the calcaneum is developed as a wide shelf. In stem otariids (*Thalassoleon*, *Hydrarctos*), the secondary shelf of the calcaneal sustentaculum is absent, developed as a thin groove or a narrow shelf.

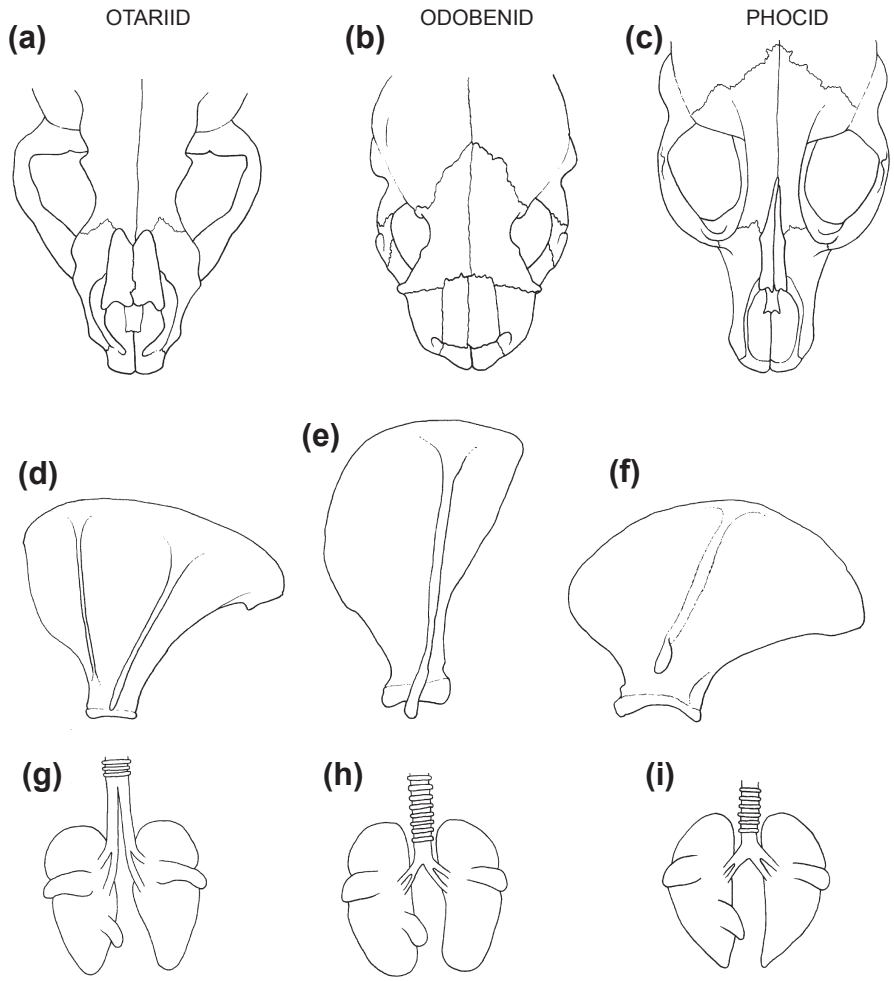

FIGURE 3.12
Otariid characters (illustrations by P. Adam). (a–c) skulls in dorsal view: frontals extend anteriorly between nasals, contact between these bones is transverse (walrus) or V-shaped (phocids) supraorbital process of the frontal bone is large and shelf-like; this process is absent in the modern walrus and phocids; (d–f) left scapulae in medial view: secondary spine subdividing the supraspinous fossa of the scapula; this ridge is absent in the walrus and phocids. (g–i) Lungs in ventral view: trachea has an anterior bifurcation of the bronchi (modified from King, 1983b). This division occurs immediately outside the lungs in the walrus and phocids.

Although the Otariidae were traditionally divided into two subfamilies, the Otariinae (sea lions) and the Arctocephalinae (fur seals) on the basis of presence (fur seals) versus absence (sea lions; see Figure 3.11(a)) of abundant underfur, there is strong agreement from morphologic and molecular data that this distinction is lacking. The fur seals are divided into two genera. *Arctocephalus* (generic name means bear head), or southern fur seals, live mostly in the southern hemisphere and a single species of northern fur seal, *Callorhinus ursinus* (generic name

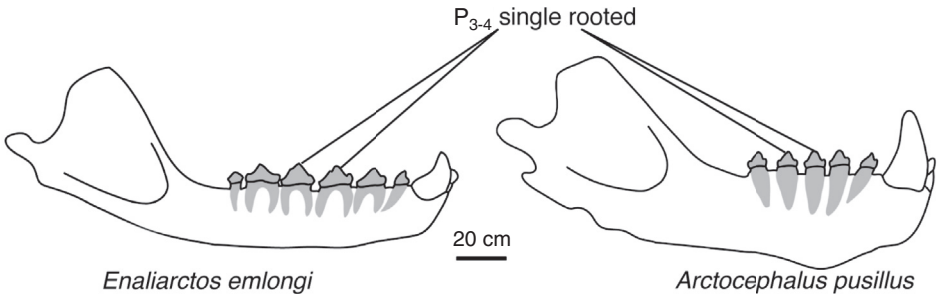

FIGURE 3.13
Crown otariid synapomorphy 10: single-rooted P_2 and P_4. From Boessenecker, 2011, with permission.

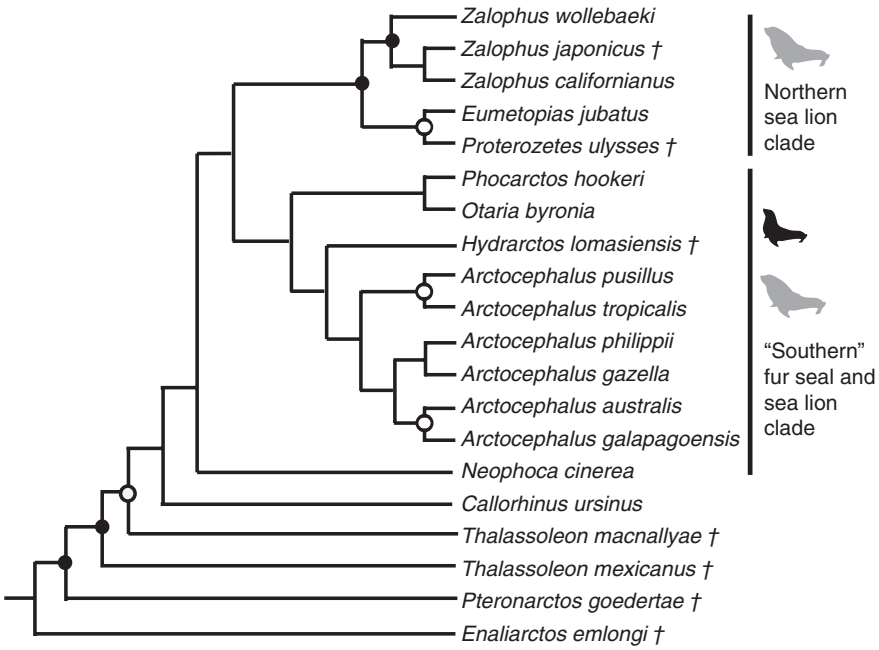

FIGURE 3.14
Phylogeny of the Otariidae based on combined data. From Churchill et al. (2014).

means beautiful nose), inhabits the northern hemisphere (see Figure 3.11(b)). Relationships among the otariids based on morphology and molecular data (Yonezawa et al. 2009; Churchill et al. 2014; Figure 3.14) suggest that *Callorhinus* is the sister group to all remaining species.

Molecular sequence data (e.g., Wynen et al. 2001; Yonezawa et al. 2009) has consistently suggested paraphyly among both fur seals (Arctocephalinae) and sea lions (Otariinae). However, phylogenetic relationships are controversial, except for the basal position of the northern fur seal (*C. ursinus*). A combined evidence analysis supports a northern sea lion clade (*Zalophus*, *Eumetopias*) as the sister

FIGURE 3.15
Life restoration of stem otariid *Pithanotaria starri*. Illustrated by Robert Boessenecker.

group to a southern clade (*Phocarctos*, *Otaria*, *Arctocephalus*, and *Arctophoca*). A novel finding of this study is recovery of a southern fur seal clade (*Arctocephalus* s.l.) (Churchill et al. 2014).

In addition to the extant fur seal genera *Callorhinus* and *Arctocephalus*, several extinct otariids are known. The earliest stem otariid is *Pithanotaria starri* (Figure 3.15) from the late Miocene (11 Ma) of California. It is a small animal characterized by double-rooted cheek teeth and a postcranial skeleton that allies it with other otariids. A second extinct stem otariid is the late Miocene taxon (8–6 Ma), *Thalassoleon* (Figure 3.16), recently reviewed by Deméré and Berta (2005), is represented by three species: *Thalassoleon mexicanus* from Cedros Island, Baja California, Mexico, and southern California; *T. macnallyae* from California; and *T. inouei* from central Japan. *Thalassoleon* is distinguished from *Pithanotaria* by its larger size and in lacking a thickened ridge of tooth enamel at the base of the third upper incisor (Berta, 1994). *Thalassoleon* spp. was found to be a paraphyletic grade of stem otariid taxa. *Hydrarctos* known from the Pliocene–Pleistocene of Peru and originally described by Muizon (1978) as a subgenus of *Arctocephalus* s.l. was recovered as the sister taxon to crown Otariidae (Churchill et al. 2014). An extinct species of northern fur seal, *Callorhinus gilmorei*, from the middle–late Pliocene, found in California and Mexico (Berta and Deméré, 1986; Boessenecker, 2011) and Japan (Kohno and Yanagisawa, 1997) has been described on the basis of lower jaws, teeth, and postcranial bones. New fossils representing a later diverging late Pliocene–early Pleistocene taxon *Callorhinus* sp. were described from northern California (Boessenecker, 2011). Several species of the southern fur seal genus *Arctocephalus* are known from the fossil record with an age range of 2.7–5 Ma (see references cited in Churchill et al. 2014).

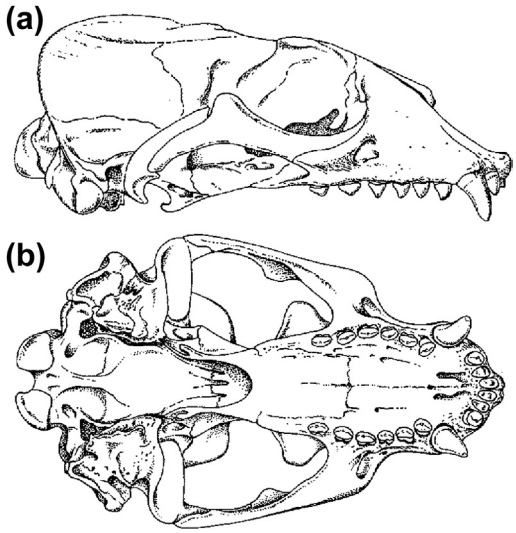

FIGURE 3.16
Skull of an early otariid, *Thalassoleon mexicanus*, from the late Miocene of western North America in (a) lateral and (b) ventral views. Original 25 cm long. From Repenning and Tedford (1977).

The fossil record of sea lions is not well known. In addition to *Proterozetes ulysses* described from the late Pliocene of Oregon (Barnes and Koretsky, 2006), there is the late Pleistocene occurrences of *Otaria byronia* from Brazil (Drehmer and Ribeiro, 1998) and *Neophoca palatina* (King, 1983b) from New Zealand (Deméré et al. 2003).

3.2.4.2 FAMILY ODOBENIDAE: WALRUSES

Arguably, the most characteristic feature of the modern walrus, *Odobenus rosmarus*, is a pair of elongated upper canine teeth (**tusks**) found in both sexes (Figure 3.11(b)). A rapidly improving fossil record indicates that these unique structures evolved in a single lineage of walruses and that "tusks do not a walrus make." The modern walrus is a large-bodied shallow diver that feeds principally on benthic invertebrates, especially mollusks. Two subspecies of *O. rosmarus* are usually recognized, *Odobenus r. rosmarus* from the North Atlantic and *O. r. divergens* from the North Pacific. A third purported subspecies, *O. r. laptevi* from the Laptev Sea, recognized by Chapskii (1940), was invalidated by Lindqvist et al. (2009), who concluded that it should be recognized as the westernmost population of the Pacific walrus. Monophyly of the walrus family, the Odobenidae, is based on five unequivocal synapomorphies (Boessenecker and Churchill, 2013, Figures 3.2, 3.17, and 3.18):

13. *Anterior narial opening large, thick-margined, and dorsoventrally elliptical.* The bony nares are primitively large and round in anterior aspect with a thin ventral margin in pinnipedimorphs and terrestrial carnivores. In odobenids, the bony nares are transversely compressed and elliptical in anterior

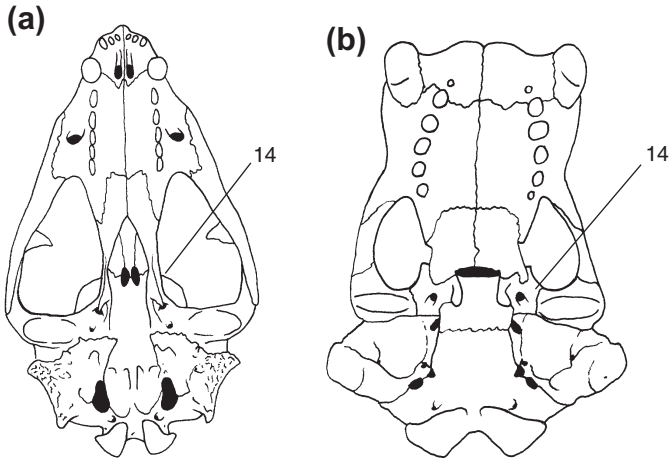

FIGURE 3.17
A walrus synapomorphy. Skull of (a) an otariid and (b) a walrus in ventral view illustrating differences in the pterygoid region. Character number: 14 = broad, thick pterygoid strut; in the otariid the pterygoid strut is narrow. From Deméré and Berta (2001).

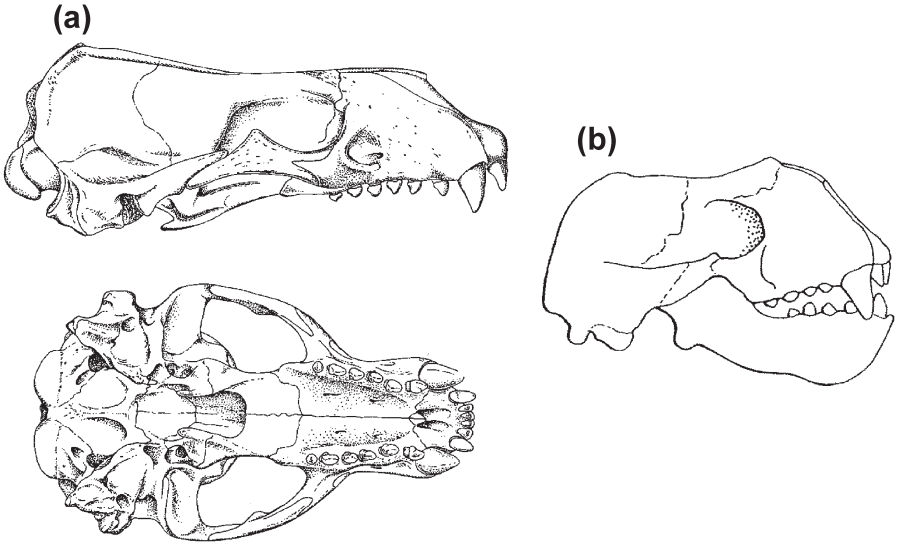

FIGURE 3.18
Skulls of fossil odobenids: (a) lateral and ventral views of *Imagotaria downsi* from the Miocene of western North America. Original 30 cm long (from Repenning and Tedford, 1977) and (b) lateral view of *Protodobenus japonicus* from the early Pliocene of Japan. Original 25 cm (from Horikawa, 1995).

aspect, and separated from the alveolar margin by a thick alveolar process of the maxilla.

14. *Pterygoid strut laterally broad and dorsoventrally thick.* The pterygoid strut is the horizontally positioned expanse of palatine, alisphenoid, and pterygoid

bones, lateral to the internal nares and hamular process. Basal pinnipedimorphs are characterized by having a narrow pterygoid strut, which in walruses is broad with a ventral exposure of the alisphenoid and pterygoid.

15. *Large epitympanic recess.* The epitympanic recess is the term given to the dorsal portion of the tympanic cavity. It is primitively small in the early pinnipedimorphs *Enaliarctos* and *Pteronarctos*, as well as otariids (*Callorhinus*). An enlarged epitympanic recess is present in all odobenids.
16. *Bony tentorium appressed to petrosal.* The bony tentorium is a subhorizontal ridge on the inside of the braincase that is associated with the meninges; it separates the cerebral hemispheres (dorsally) from the cerebellum (ventrally). It is primitively separated from the dorsal surface of the petrosal, a condition present in enaliarctines, *Callorhinus*, and Desmatophocidae. In modern *Odobenus* and all fossil odobenids, the bony tentorium is closely appressed to the dorsal surface of the petrosal.
17. *3-rooted* M^1. The primitive condition characterizes the large molars of terrestrial carnivores and is present in early diverging odobenids including *Prototaria*, *Proneotherium*, *Neotherium*, and *Kamtschatarctos*. A double-rooted M^1 also appears in *Pseudotaria*, *Imagotaria*, *Pontolis*, fossil Otariidae, and most Phocidae. A single-rooted M^1 occurs in extant Otariidae, *Mirounga*, *Dusignathus seftoni*, *Gomphotaria*, *Aivukus*, *Protodobenus*, and *Ontocetus*. The M^1 is absent in *Odobenus* and *Valenictus*.

Morphological study of evolutionary relationships among walruses has identified stem "imagotariine" walruses and two monophyletic groups of crown Odobenidae. At the base of walrus evolution are the stem walruses "imagotariines" *Proneotherium* and *Prototaria*, from the middle Miocene (16–14 Ma) of the North Pacific (Deméré and Berta, 2001; Kohno, 2006). Other stem "imagotariine" odobenids include *Neotherium*, *Imagotaria* (Figure 3.18), *Pseudotaria*, and *Pelagiarctos* (Figure 3.19) from the middle–late Miocene of the eastern North Pacific (Kohno, 2006; Boessenecker and Churchill, 2013). These archaic walruses are characterized by unenlarged canines and narrow, multiple-rooted premolars with a trend toward molarization, adaptations suggesting retention of the fish diet hypothesized for archaic pinnipeds rather than the evolution of the specialized mollusk diet of the modern walrus. The Dusignathinae includes the extinct genera *Dusignathus* and *Gomphotaria*. The Odobeninae includes in addition to the modern walrus, *Odobenus*, the extinct genera *Aivukus*, *Ontocetus* (=*Prorosmarus*, *Alachterium* fide Kohno and Ray, 2008), *Protodobenus* and *Valenictus* (Boessenecker and Churchill, 2013). Dusignathine walruses developed enlarged upper and lower canines, whereas odobenines evolved only the enlarged upper canines seen in the modern walrus.

The dusignathine walrus, *Dusignathus santacruzensis*, and the odobenine walrus, *Aivukus cedroensis*, first appeared in the late Miocene of California and Baja California, Mexico. Early diverging odobenine walruses are now known from both sides of the Pacific and Atlantic (i.e., *Ontocetus emmoni*) in the early Pliocene (Ray and Kohno, 2008). The most completely known fossil odobenine walrus, *Valenictus chulavistensis*, was described by Deméré (1994b) as being closely

FIGURE 3.19
Life restoration of the head of the "imagotariine" walrus *Pelagiarctos* sp. Illustrated by Robert Boessenecker.

related to modern *Odobenus*, but distinguished from it in having no teeth in the lower jaw and lacking all upper postcanine teeth. The toothlessness (except for tusks) of *Valenictus* is unique among pinnipeds but parallels the condition seen in modern suction-feeding beaked whales and the narwhal.

Remains of the modern walrus *Odobenus* date back to the early Pliocene of Belgium; this taxon appeared approximately 600,000 years ago in the Pacific.

3.2.4.3 FAMILY PHOCIDAE: SEALS

The second major grouping of living seals, the phocids, are often referred to as the earless seals because they lack of visible ear pinnae (external ear flap), a characteristic that readily distinguishes them from otariids. Another characteristic phocid feature is their method of movement on land. The phocids are unable to turn their hindflippers forward so they move on land by undulations of the body (described in more detail in Chapter 8). Other characteristics of phocids include their larger body size in comparison to otariids, averaging as much as 2 tons in elephant seal males. Several phocids, most notably the elephant seals and the Weddell seal, are spectacular divers that feed on pelagic, vertically migrating squid and fish at depths of 1000 m or more.

Among characters Wyss (1988) used to support monophyly of the Family Phocidae confirmed by Amson and Muizon (2014) in the context of intraphocid relationships (Figures 3.2, 3.20, and 3.21) are the following:

18. *Lack the ability to draw the hindlimbs forward under the body due to a massively developed astragalar process and greatly reduced calcaneal tuber*. The phocid astragalus (ankle bone) is distinguished by a strong posteriorly directed process over which the tendon of the flexor hallucis longus passes. The calcaneum (heel bone) of phocids is correspondingly modified. The

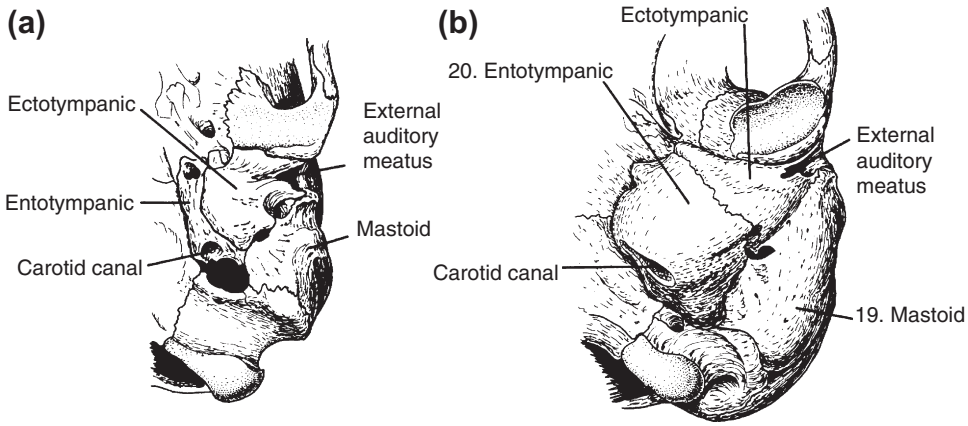

FIGURE 3.20
Phocid synapomorphies. Ventral view of ear region of (a) an otariid and (b) a phocid. Character numbers (see text for further description): 20 = pachyostotic mastoid bone—this is not the case in other pinnipeds; 21 = greatly inflated entotympanic bone—in other pinnipeds, this bone is flat or slightly inflated. Modified from King (1983b).

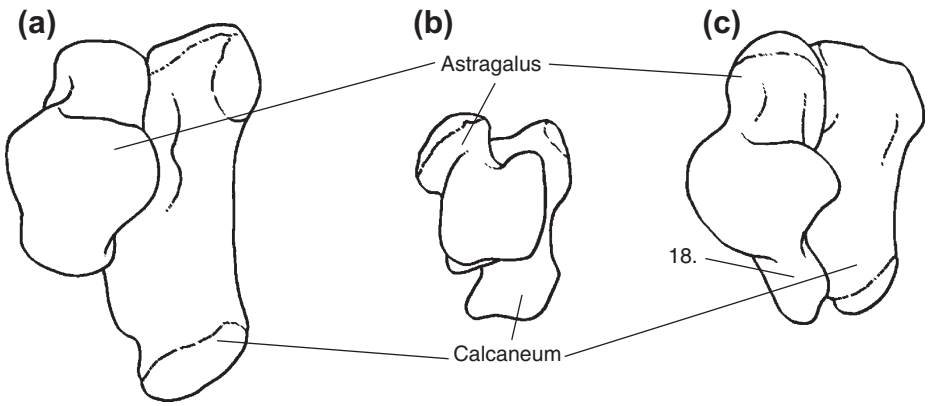

FIGURE 3.21
Phocid synapomorphies. Left astragalus (ankle) and calcaneum (heel) of (a) an otariid, (b) a walrus, and (c) a phocid. Character numbers: 19 = lack the ability to draw the hindlimbs under the body due to a massively developed astragalar process and greatly reduced calcaneal tuber; these modifications do not occur in other pinnipeds. Illustrations by P. Adam.

calcaneal tuber is shortened and projects only as far as the process of the astragalus. This arrangement prevents anterior flexion of the foot, resulting in an inability to bring their hindlimbs forward during locomotion on land.

19. *Pachyostic mastoid region.* In phocids, the mastoid (ear) region is composed of thick, dense bone (pachyostosis), which is not the case in otariids or the walrus.
20. *Greatly inflated entotympanic bone.* In phocids, the entotympanic bone (one of the bones forming the earbone or tympanic bulla) is inflated. In other pinnipeds, the entotympanic bone is either flat or only slightly inflated (Figure 3.20).

FIGURE 3.22
Representative monachine and phocine seals: (a) the monachine northern elephant seal (*Mirounga angustirostris*) (b); and (b) the phocine ribbon seal (*Histriophoca fasciata*). Illustrated by Carl Buell.

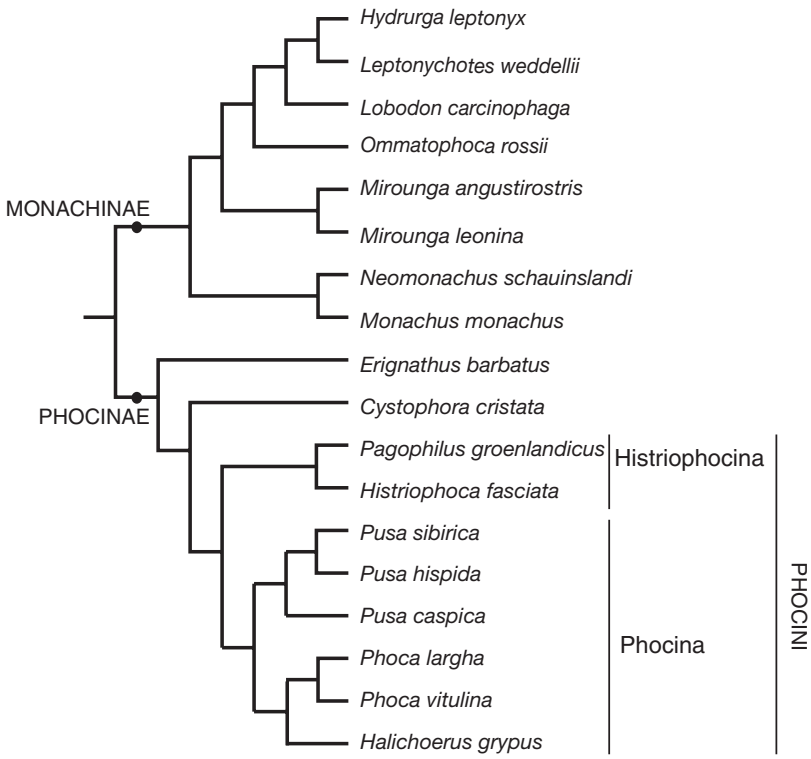

FIGURE 3.23
Phylogeny of the Phocidae based on molecular data. From Davis et al. 2004.

Most morphologic and molecular data support monophyly of two subgroups of phocids Monachinae (monk seals and southern seals including the elephant seals) and Phocinae (Northern Hemisphere seals) (Figure 3.22) (Amson and Muizon, 2014; Berta et al. 2015). In the absence of a robust phylogenetic analysis for phocids that includes extant and extinct taxa, we provide a tree based on molecular data (Figure 3.23) that has received general agreement among studies.

In addition to Davis et al. (2004) other molecular studies provide data on phocid interrelationships (e.g., (Davis et al. 2004; Higdon et al. 2007; Fulton and Strobeck, 2010; Nyakatura and Bininda-Emonds, 2012; see Figure 3.23). Among phocines, consistent strong support is found for recognition of the bearded seal (*Erignathus barbatus*) as a sister group to the remaining taxa, followed by the hooded seal (*Cystophora cristata*). Most recent studies have found support for the next branch of the tree being *Histriophoca* and *Pagophilus* (=Histriophocina) as a sister group to the remaining taxa. Of the remaining species (*Phoca*, *Pusa*, and *Halichoerus* = Phocina), there is disagreement (e.g., Arnason et al. 2006). The gray seal (*Halichoerus grypus*) is positioned as the sister species of the Caspian seal (*Pusa caspica*) in several studies (e.g., Fulton and Strobeck, 2010; Nyakatura and Bininda-Emonds, 2012) but in other studies *Halichoerus* clusters within *Phoca* (e.g., Fulton and Strobeck, 2010). The sister-group relationship between the harbor seal (*Phoca vitulina*) and the spotted seal (*Phoca largha*) is consistent and well supported. There is also strong support for three monachine lineages: (1) the Hawaiian monk seal (*Neomonachus schauinslandi*), sister taxon to the extinct Caribbean monk seal (*Neomonachus tropicalis*); (2) monophyletic elephant seals (*Mirounga* spp.) sister taxon to (3) the lobodontine seals (i.e., *Hydrurga*, *Lobodon*, *Ommatophoca*, and *Leptonychotes*) (e.g., Davis et al. 2004; Higdon et al. 2007; Scheel et al. 2014).

The report of a fossil phocid from the late Oligocene (29–23 Ma) of South Carolina (Koretsky and Sanders, 2002), if its stratigraphic provenance is correct, makes it the oldest known phocid and the oldest known pinniped (see Figure 3.1). Prior to this record, phocids were unknown until the middle Miocene (15 Ma) when both phocine (i.e., *Leptophoca lenis*) and *Leptophoca amphiatlantica;* the latter from the Antwerp Basin, Belgium (Koretsky et al. 2012) and "monachine" (i.e., *Monotherium*? *wymani*) seals became distinct lineages in the North Atlantic (Koretsky, 2001; Koretsky and Ray, 2008). Remains of an indeterminate monachine recorded from the middle Miocene of Malta (Bianucci et al. 2011) and the even older purported monachine *Afrophoca libyca* from the early–middle Miocene of Libya (Koretsky and Domning, 2014) implies a long history of monachine phocids in the Mediterranean Basin. Other fossil true seals are represented by well-preserved skeletal material. For example, *Acrophoca longirostris* (Figure 3.24) and *Piscophoca pacifica* (Muizon, 1981) have been reported from the late Miocene and early Pliocene Pisco Formation of Peru and probably Chile (Walsh and Naish, 2002), and more recently *Hadrokiris martini* (Amson and Muizon, 2014) and *Homiphoca capensis* from South Africa (Hendey and Repenning, 1972; Govender et al. 2012). Recent studies (Amson and Muizon, 2013; Berta et al. 2015) indicate that *Acrophoca*, *Piscophoca*, *Hadrokiris*, and *Homiphoca* are likely monachines. Phocines evolved in the vast inland Paratethys Sea from 14 to 10 Ma (Koretsky, 2001; Koretsky and Grigorescu, 2002; Koretsky and Holec, 2002; Koretsky and Rahmat, 2013; see also Chapter 6) and differentiated mostly in the Pleistocene.

3.2.5 Walruses and Phocids Linked?

As noted previously, an alliance between the walrus and phocid seals and their extinct relatives, the Phocomorpha clade, has been proposed based on

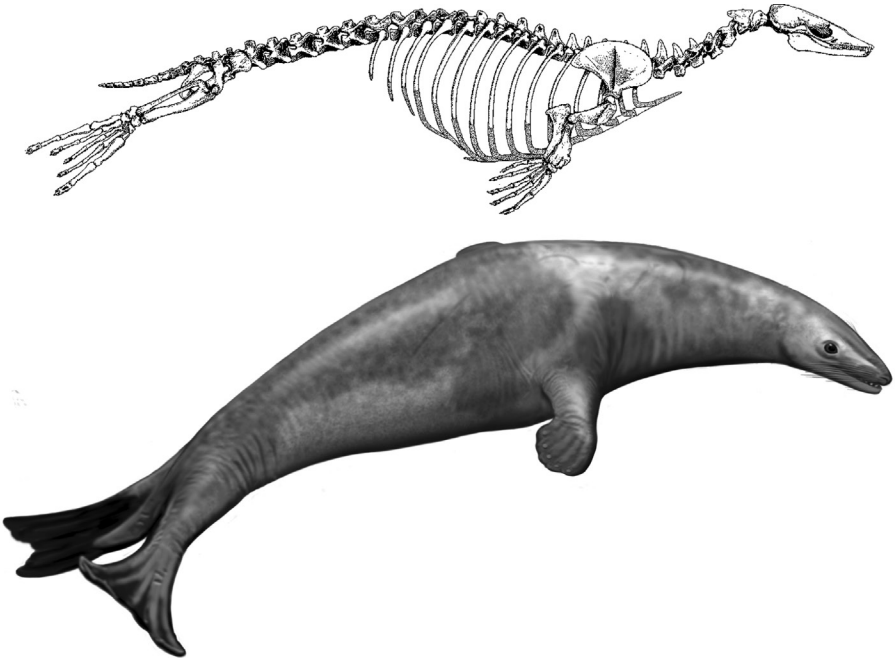

FIGURE 3.24
Skeleton of an archaic phocid, *Acrophoca longirostris*, from the Miocene of Peru (from Muizon, 1981) and life restoration. Illustrated by Carl Buell.

morphologic data. Among the synapomorphies that unite these pinnipeds are the following (see Figure 3.2):

21. *Middle ear bones enlarged.* The middle ear bones of the walrus and phocids are large relative to their body size, which is not the case in otariids and terrestrial carnivores.
22. *Abdominal testes.* The testes are abdominal (inguinal) in phocids and the walrus. In contrast, the testes of otariids and terrestrial carnivores lie outside the inguinal ring in a scrotal sac.
23. *Primary hair nonmedullated.* The outer guard hairs in the walrus and phocids lack a pith or medulla, which is present in otariids and other carnivores (see Chapter 7).
24. *Thick subcutaneous fat.* The walrus and phocids are characterized by a thick blubber layer; this subdermal layer is less well developed in otariids and terrestrial carnivores.
25. *External ear pinna lacking.* The walrus and phocids lack external ear pinnae, the presence of which characterizes otariids and terrestrial carnivores.
26. *Venous system with inflated hepatic sinus, well-developed caval sphincter, large intervertebral sphincter, duplicate posterior vena cava, and gluteal route for hindlimbs.* The walrus and phocids share a specialized venous system that is related, in part, to their exceptional diving capabilities (see Chapter 10). In contrast, otariids and terrestrial carnivores have a less specialized venous system that more closely approximates the typical terrestrial mammalian pattern.

The question of whether walruses are more closely related to phocids or to otariids involves further exploration of both morphological and molecular data sets. Study of basal walruses will likely provide additional characters that can be compared for alliance with either otariids or phocids. The molecular results, which consistently support a link between otariids and the walrus, have been explained as a methodological problem or a "long-branch attraction effect." When dealing with sequence data, branch lengths refer to the expected amount of evolutionary change along that branch. The walrus lineage is a relatively long branch and it is unlikely that intraspecific variation among the walrus subspecies will aid in bisecting the branch, although this has yet to be examined. Because there is a tendency for longer branches to attract one another and thus give a misleading tree, the walrus and otariid alliance may be incorrect. A more conservative interpretation of the molecular data is that the walrus is an early, but not first, independent divergence from the common pinniped ancestor (Lento et al. 1995).

3.2.6 Desmatophocids: Phocid Relatives or Otarioids?

Study of pinniped evolutionary relationships has identified a group of fossil pinnipeds including *Desmatophoca* and *Allodesmus* (Figure 3.25) that are positioned as the common ancestors of phocid pinnipeds (Berta and Wyss, 1994). This interpretation differs from previous work that recognized

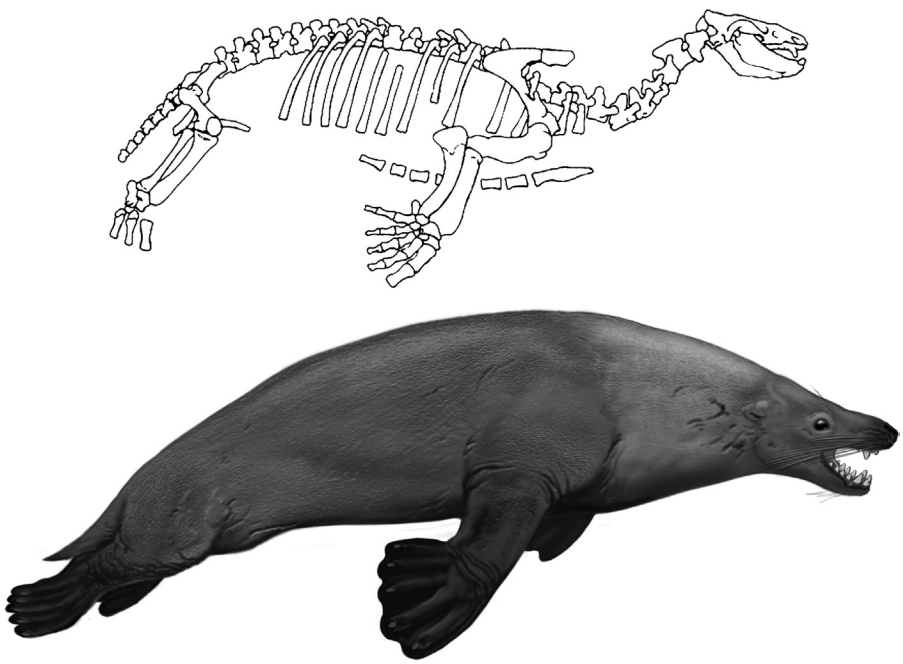

FIGURE 3.25
Skeleton of the desmatophocid, *Allodesmus kernensis*, from the Miocene of western North America. Original 2.2 m long (from Mitchell, 1975) and life restoration. Illustrated by Carl Buell.

desmatophocids as otarioid pinnipeds, a grouping that includes walruses (Barnes, 1989). The question of otarioid monophyly was examined in a comprehensive pinniped data set. The strict consensus tree that resulted by forcing otarioid monophyly was 34 steps longer than the preferred hypothesis (Berta and Wyss, 1994).

Desmatophocids are known from the early middle Miocene (23–15 Ma) of the western US and Japan. Reported occurrences of *Desmatophoca* from Oregon confirm the presence of sexual dimorphism and large body size in these pinnipeds (Deméré and Berta, 2002). Allodesmids are known from the middle to late Miocene of California and more recently from Japan (Barnes and Hirota, 1995). They are a diverse group characterized by, among other characters, pronounced sexual dimorphism, large eye orbits, bulbous cheek tooth crowns, and deep lower jaws.

A number of features are shared among phocids and their close fossil relatives *Allodesmus* and *Desmatophoca* (identified as the Phocoidea clade) and hence support a close link between these taxa (see Figure 3.2). These synapomorphies include among others:

27. *Posterior termination of nasals posterior to contact between the frontal and maxilla bones.* In phocids and desmatophocids, the V-shaped contact between the frontal and maxilla bones is the result of the nasals extending posteriorly between these bones (see Figure 3.12).
28. *Squamosal jugal contact mortised.* A mortised or interlocking contact between the squamosal and jugal (cheekbones) distinguishes phocids and desmatophocids from other pinnipeds in which these bones overlap one another in a splint-like arrangement (see Figure 3.3).

3.3 SUMMARY AND CONCLUSIONS

Pinniped monophyly is a consistently well-supported hypothesis based on both morphological and molecular data. The closest relatives of pinnipeds are arctoid carnivores, with most evidence supporting either a link between pinnipeds and ursids or pinnipeds and mustelids.

The earliest stem taxa, pinnipedimorphs (i.e., *Enaliarctos*) appear in the fossil record approximately 27–25 Ma in the North Pacific. Crown pinniped lineages diverged shortly thereafter, with the first appearance of phocid seals in the North Atlantic. Phocids are divided into two monophyletic subgroups, southern monachines and northern phocines. Walruses appeared about 10 million years later in the North Pacific. A rapidly improving fossil record indicates that enlarged tusks, characteristic of both sexes of the modern walrus, were not present ancestrally in walruses. Two monophyletic lineages of walruses are recognized (Dusignathinae and Odobeninae). The last pinniped lineage to appear in the fossil record, the otariids, is only known as far back as 11 Ma in the North Pacific. Neither sea lions (Otariinae) nor fur seals (Arctocephalinae) are monophyletic.

A remaining conflict is the precise position of the walrus. Most morphologic data support a phocid and walrus alliance, whereas molecular and combined data consistently support uniting otariids and the walrus. Resolution of these conflicts will likely benefit from a detailed exploration of both morphologic and molecular data sets.

3.4 FURTHER READING AND RESOURCES

Relationships among various arctoid carnivores and pinnipeds are reviewed in Flynn and Wesley-Hunt (2005), Nyakatura and Bininda-Emonds (2012). See Berta (1991, 1994) and Barnes (1989, 1992) for stem pinnipedimorphs, Repenning and Tedford (1977) and Churchill et al. (2014) for fossil otariids and Boessenecker and Churchill (2013) walruses, and Muizon (1981) and Koretsky (2001), Koretsky and Ray (2008), and Amson and Muizon (2014) for fossil phocids. Reviews of the evolution and phylogeny of walruses include Deméré (1994a,b), Kohno (2006), Deméré and Berta (2001), and Boessenecker and Churchill (2013). For alternative views on phocid phylogeny, see the morphologically based studies of Wyss (1988), Bininda-Emonds and Russell (1996), Bininda-Emonds et al. (1999), and Amson and Muizon (2014) and the molecular studies of Arnason et al. (2006), Higdon et al. (2007), and Fulton and Strobeck (2010). Yonezawa et al. (2009) provides molecular evidence for otariid relationships, but also see Churchill et al. 2014 for a different view based on combined data.

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